ILKIN ISLER, Ph.D.

ilkinisler@gmail.com LinkedIn Google Scholar ilkinisler.com

PROFESSIONAL SUMMARY

Senior AI Engineer working in a forward-deployed role: meets with business teams, scopes complex business needs, and builds the AI solution from first prototype to trusted production. Builds models, not just uses them: Ph.D. in Computer Science focused on designing novel neural network architectures, with hands-on work across computer vision, NLP, agentic systems, automation, and dataset creation. Works across the full LLM stack: architecture fundamentals, pre-training and fine-tuning, evaluation, RAG over vector databases, serving and inference optimization for speed and cost, and securing LLM applications. Independently built MCPGates, the organization's enterprise AI platform (20+ production applications), and its first RAG and evaluation systems; leads a five-person team across AI infrastructure, Finance, and Estimating & Budgeting.

TECHNICAL SKILLS

LLM Foundations: Transformer architecture (tokenization, attention, sampling), prompt engineering, LLM serving and inference optimization (latency and cost), LLM security
Retrieval & Knowledge Graphs: RAG (vector, hybrid BM25 + vector, and GraphRAG), temporal knowledge graphs (Graphiti-style, built from scratch), data and document ingestion
Agents: Multi-agent systems, agent orchestration, MCP servers, function calling, agent skills, human-in-the-loop systems
Evaluation & Trustworthy AI: Hallucination detection, groundedness evaluation, citation verification, Vectara HHEM, LLM-as-judge, RAGAS, DeepEval, MRR, benchmark and synthetic data creation, regression analysis, uncertainty quantification, explainable AI
Machine Learning & Deep Learning: PyTorch, MONAI; vision transformers (Swin UNETR), CNNs (U-Net), GANs, LSTMs, gradient boosting (XGBoost); multi-GPU distributed training (DDP), neural network architecture design, pre-training and fine-tuning (self-supervised, contrastive, transfer learning), medical image segmentation
Languages, Data & Tools: Python (pandas, NumPy, scikit-learn, spaCy), SQL, FastAPI, LlamaIndex, pgvector (Postgres), Neo4j, SQLite FTS5, FAISS, RBAC/ABAC

PROFESSIONAL EXPERIENCE

UNIVERSAL CREATIVE — *Orlando, FL*

Senior AI Engineer, *AI Strategy & Solutions* — Aug 2025 – Present

- Independently architected, built, and deployed MCPGates, the in-house enterprise AI platform that hosts the team's 20+ production AI applications in one place. New applications ship faster because they reuse shared core components (retrieval and reinforcement learning pipelines, service-account connections to external data sources, MCP servers, tool calling, and agent skills) and inherit one Microsoft Entra ID single sign-on (SSO), role-based access control (RBAC), and a cybersecurity-approved architecture instead of repeating security approval for each one. Serves 200–300 users across multiple business units, with Application Insights monitoring and logging dashboards.
- Lead the Finance AI initiative as solution architect and responsible owner: designed from scratch three human-in-the-loop applications for reporting and forecasting to replace a frequently reworked, file-per-project spreadsheet process, and lead the developers building them. Cash Flow & Earned Value is in production with 70+ finance users globally, unifying finance data in one place and saving users 15–30 minutes per analysis and about an hour per forecast; Monthly Update Insights (top cost drivers each month) and Risk & Contingency (transaction risk flags) have their first versions in production, with more capabilities in development.
- Independently architected, built, and deployed Universal Creative's first LLM and RAG evaluation pipelines, scoring hallucination, groundedness, citation correctness, and factual consistency. They gate new methods before production, monitor live chatbots with hallucination scores, and select the best grounded answer at runtime (or decline to answer when none is grounded) across four production applications, including Specs Q&A and Pricing Intelligence.
- Lead the Estimating & Budgeting AI initiative as solution architect and responsible owner: designed from scratch the Estimator Builder, an end-to-end, human-in-the-loop suite of 12 purpose-built AI applications for ~80 users, and lead the developers building it, with 6+ shipped. It includes Drawing Intelligence (reads design drawings and generates scope), Concept Estimate Automation (converts messy, non-standardized show element lists into structured estimates and tracks how they change), and Pricing Intelligence (retrieves everything previously paid for any item from more than 2 million files). Cut estimating from days to minutes and made it traceable and consistent.
- Independently architected, built, and deployed a project-intelligence application that shows senior leadership what changed across a project and how it affects budgets, plans, and schedules. Built on a temporal knowledge graph (similar to Graphiti, using Neo4j): project decks and updates from design, scheduling, estimating, budgeting, and finance systems are ingested as episodes, LLMs extract entities and relationships into a graph that keeps every fact's history, and hybrid semantic, keyword, and graph-traversal retrieval answers questions about them.
- Extended and improved the in-house data ingestion pipeline to make all structured and unstructured project data searchable for the project-intelligence application, from invoices, RFIs, change orders, and issues to PRDs, meeting notes, presentations, emails, and screenshots.
- Lead the five-person AI engineering team as technical lead across the infrastructure, finance, and cost-estimating workstreams, delivering on scope, schedule, and budget; co-authored the AI engineering guidance (build and deployment, retrieval, hallucination detection, evaluation, data security) that external vendors must meet, ran two AI lunch-and-learns (~100 attendees), and advised senior leaders on AI adoption in 5–6 working sessions.

AI Software Engineer, *Engineering & Safety* — Jan 2025 – Aug 2025

- Independently architected, built, and deployed the company's first retrieval-enabled chatbot for Engineering & Safety over engineering specifications, with RAG, hallucination detection, groundedness scoring, citation verification, topic modeling, metadata tagging, and structured extraction. Evaluated each component on self-built benchmark datasets, including synthetic question-answer data, using MRR for retrieval and RAGAS, DeepEval, BERTScore, ROUGE, and LLM-as-judge for factual consistency and response quality. This work became the core of MCPGates and its production evaluation pipelines.

Northwestern University, Department of Radiology — *Chicago, IL*

Doctoral Researcher, *Machine and Hybrid Intelligence Lab* — May 2023 – Dec 2024

- Led collaborations across multidisciplinary teams to improve medical image visualization tools with detection and segmentation capabilities for oncology and move AI models closer to real-world clinical deployment.
- Managed project timelines and deliverables, and wrote the technical documentation supporting AI model development and interpretation for clinical use.

University of Central Florida, College of Engineering and Computer Science — *Orlando, FL*

Graduate Teaching Assistant — Aug 2023 – Dec 2024

- Assisted with Advanced Artificial Intelligence and Advanced Computer Networks courses, and taught ~4 lectures.

Graduate Research Assistant — Jan 2021 – Aug 2023

- Designed novel 3D neural network architectures and compound loss functions combining overlap and boundary-distance objectives for tumor and organ-at-risk segmentation, trained with multi-GPU distributed training (PyTorch DDP) on ~1,000 3D CT and MRI volumes merged and harmonized from multiple datasets, expanded with extensive data augmentation; self-supervised and uncertainty-guided coarse-to-fine approaches achieved state-of-the-art results at publication (IEEE ICECCME 2023, IEEE ACDSA 2025), including raising the transformer-based Swin UNETR's Dice from 0.47 to 0.64, and an improved U-Net reached ~80% vs. 63% Dice over baseline (SPIE Medical Imaging 2022).
- Predicted cancer patient survival from radiomics and hand-crafted imaging features.
- Developed uncertainty quantification, explainability, and anatomy-aware post-processing methods that guided the model to refine its uncertain regions, improving accuracy, and gave clinicians uncertainty estimates showing when to rely on its output; validated reliability with a broad set of segmentation metrics, including Dice and HD95, alongside confidence and failure analysis.
- Co-authored a multi-institutional, five-center pancreas segmentation study (767 MRI scans from 499 participants, plus CT from 1,350 patients) published in Medical Image Analysis.

Earlier Experience

- **Python Developer (Contract), Universal Orlando Resort** (Orlando, 2022): resolved 100+ vulnerabilities in cloud data pipelines and optimized ML workflows with Azure Databricks and MLflow.
- **Data Analyst, Simsoft Information Technologies** (Ankara, 2020): built LSTM models for anomaly detection and predictive maintenance on industrial sensor data; modeled COVID-19 spread.
- **Research Intern, UCF Synthetic Reality Lab** (Orlando, 2019): applied GANs to facial expression recognition and human-to-avatar translation.
- **Data Analyst Intern, Carbon Health** (San Francisco, 2018): built predictive models that cut clinic wait times by 30%.

EDUCATION

Ph.D., Computer Science, University of Central Florida, Orlando, FL — Aug 2025
Dissertation: *Advanced Deep Learning Algorithms for Medical Imaging Applications: Tumor and Organ at Risk Segmentation*

MS, Computer Science, University of Central Florida, Orlando, FL — Dec 2022

BS, Computer Science, Hacettepe University, Ankara, Turkey — June 2020

PUBLICATIONS & ACADEMIC SERVICE

- Zhang, Z., …, **Isler, I.**, et al. Large-scale multi-center CT and MRI segmentation of pancreas with deep learning. *Medical Image Analysis*, 2025.
- **Isler, I.**, Mohaisen, D., Lisle, C., Turgut, D., Bagci, U. Uncertainty-Guided Coarse-to-Fine Tumor Segmentation with Anatomy-Aware Post-Processing. *IEEE ACDSA*, 2025.
- Jha, D., …, **Isler, I.**, et al. Practical and Ethical Considerations for Generative AI in Medical Imaging. *MICCAI Workshop*, 2024.
- **Isler, I.**, Jha, D., Lisle, C., et al. Self-Supervised Learning for Organs At Risk and Tumor Segmentation with Uncertainty Quantification. *IEEE ICECCME*, 2023.
- **Isler, I.**, Lisle, C., Rineer, J., et al. Enhancing Organ at Risk Segmentation with Improved Deep Neural Networks. *SPIE Medical Imaging*, 2022.

Peer Reviewer: Medical Physics; IEEE Journal of Biomedical and Health Informatics; IEEE Transactions on Medical Imaging

ADDITIONAL

Project: ilkinisler.com, a personal site with a grounded AI assistant: hybrid BM25 + embedding retrieval, LLM answers scored for groundedness with Vectara's HHEM model (declining below a confidence threshold), and a FastAPI backend serving LLM inference on Google Cloud Run.

Media: Featured in UCF News, *The Mind to Move Mountains* (2026).

Athletics: Competitive powerlifter: four-time USA Powerlifting Florida state record holder, two-time Overall Champion of Türkiye, 1st Place Overall at the IPSU European Championship